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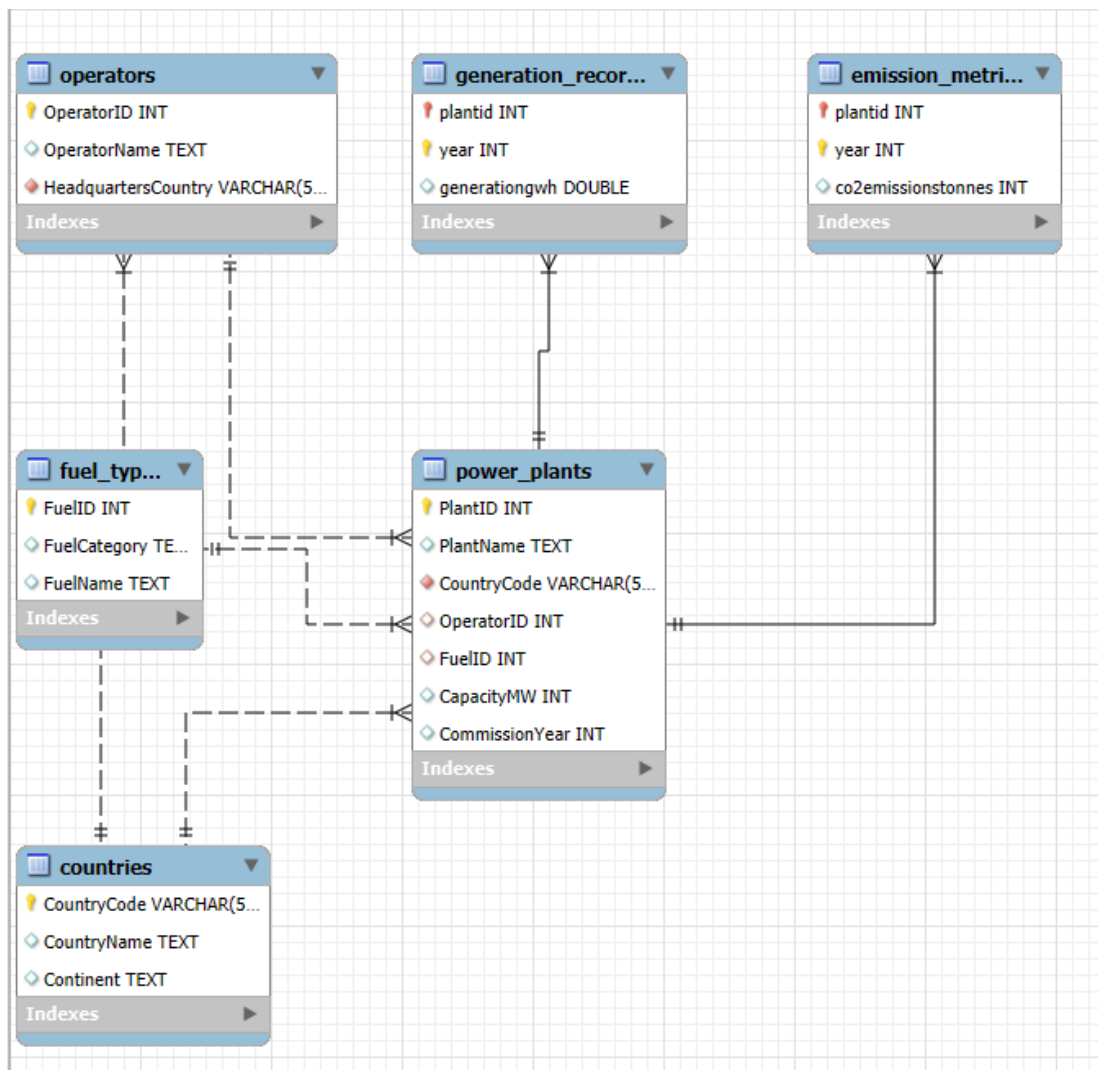
CSC 621 Database Management

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Assignment 3

- EER Diagram made in MySQL



SQL Query Code:

-- 1.

```
select power_plants.PlantName, countries.CountryName, operators.OperatorName, fuel_types.FuelCategory,
fuel_types.FuelName, power_plants.CapacityMW, power_plants.CommissionYear from power_plants
join countries on power_plants.CountryCode = countries.CountryCode
join operators on power_plants.OperatorID = operators.OperatorID
join fuel_types on power_plants.FuelID = fuel_types.FuelID
order by power_plants.CapacityMW desc;
```

	PlantName	CountryName	OperatorName	FuelCategory	FuelName	CapacityMW	CommissionYear
▶	Three Gorges	China	China Yangtze Power	Renewable	Hydro	22500	2003
	Itaipu	Brazil	Electrobras	Renewable	Hydro	14000	1984
	Tuoketuo	China	China Yangtze Power	Fossil	Coal	13200	1995
	Kashiwazaki-Kariwa	Japan	Tepco Energy	Nuclear	Uranium	8212	1985
	Grand Coulee	United States	Global Hydro Inc	Renewable	Hydro	6809	1942
	Vindhyachal	India	NTPC Limited	Fossil	Coal	4760	1987
	Janschwalde	Germany	Rhein Power AG	Fossil	Coal	3000	1981
	Loy Yang	Australia	Snowy Hydro	Fossil	Coal	2210	1984

-- 2.

```
select power_plants.PlantName, power_plants.CountryCode, generation_records.year,
generation_records.generationgwh from power_plants
join generation_records on power_plants.PlantID = generation_records.plantid
where generation_records.year = 2024
order by generation_records.generationgwh desc;
```

	PlantName	CountryCode	year	generationgwh
▶	Three Gorges	CHN	2024	98800
	Itaipu	BRA	2024	9000.1
	Tuoketuo	CHN	2024	71000.3
	Vindhyachal	IND	2024	35600.8
	Grand Coulee	USA	2024	21500.5
	Janschwalde	DEU	2024	18400.2
	Loy Yang	AUS	2024	14200.4
	Kashiwazaki-Kariwa	JPN	2024	0

-- 3.

```
select power_plants.PlantName, power_plants.CountryCode, generation_records.year,
generation_records.generationgwh, emission_metrics.co2emissionstonnes from power_plants
join generation_records on power_plants.PlantID = generation_records.plantid
```

join emission_metrics on power_plants.PlantID = emission_metrics.plantid

where generation_records.year = 2024

order by emission_metrics.co2emissionstonnes asc;

	PlantName	CountryCode	year	generationgwh	co2emissionstonnes
▶	Grand Coulee	USA	2024	21500.5	0
	Itaipu	BRA	2024	89000.1	0
	Kashiwazaki-Kariwa	JPN	2024	0	0
	Three Gorges	CHN	2024	98800	0
	Loy Yang	AUS	2024	14200.4	1950000
	Janschwalde	DEU	2024	18400.2	2450000
	Vindhyachal	IND	2024	35600.8	3100000
	Tuocketuo	CHN	2024	71000.3	5200000

-- 4.

with total_generationgwh as (

select operators.OperatorID as oID, SUM(generation_records.generationgwh) as totalG

from generation_records

right join power_plants on generation_records.plantID = power_plants.PlantID

right join operators on power_plants.OperatorID = operators.OperatorID

where generation_records.generationgwh IS NOT NULL

group by operators.OperatorID

)

select distinct operators.OperatorName, operators.HeadquartersCountry, total_generationgwh.totalG

from total_generationgwh

join operators on total_generationgwh.oID = operators.OperatorID

join power_plants on operators.OperatorID = power_plants.OperatorID

join generation_records on power_plants.PlantID = generation_records.plantid

order by total_generationgwh.totalG desc;

	OperatorName	HeadquartersCountry	totalG
▶	China Yangtze Power	CHN	169800.3
	Electrobras	BRA	89000.1
	NTPC Limited	IND	35600.8
	Global Hydro Inc	USA	21500.5
	Rhein Power AG	DEU	18400.2
	Snowy Hydro	AUS	14200.4
	Tepco Energy	JPN	0

-- 5.

```
with total_country_power as (  
select countries.CountryCode as country, SUM(generation_records.generationgwh) as country_total  
from generation_records  
right join power_plants on generation_records.plantID = power_plants.PlantID  
right join countries on power_plants.CountryCode = countries.CountryCode  
where generation_records.generationgwh IS NOT NULL  
group by countries.CountryCode  
,  
total_country_emissions as (  
select countries.CountryCode as country, SUM(emission_metrics.co2emissionstonnes) as country_total  
from emission_metrics  
right join power_plants on emission_metrics.plantID = power_plants.PlantID  
right join countries on power_plants.CountryCode = countries.CountryCode  
where emission_metrics.co2emissionstonnes IS NOT NULL  
group by countries.CountryCode  
)  
select countries.CountryName, total_country_power.country_total, total_country_emissions.country_total  
from countries  
join total_country_power on countries.CountryCode = total_country_power.country  
join total_country_emissions on countries.CountryCode = total_country_emissions.country  
order by total_country_power.country_total desc;
```

	CountryName	country_total	country_total
▶	China	169800.3	5200000
	Brazil	89000.1	0
	India	35600.8	3100000
	United States	21500.5	0
	Germany	18400.2	2450000
	Australia	14200.4	1950000
	Japan	0	0